

Lesson 2 · about 60 minutes · everyone builds; no electrical hazard

What you need

- Two bar magnets, one horseshoe magnet
- Iron filings in a shaker (or steel wool cut fine)
- A sheet of clear acrylic or stiff card
- Several small compasses

- Castor or mineral oil in a shallow clear dish
- Grass seed or fine semolina
- The electroscope from Lesson 1, and a charged comb

Predict first

You will lay a card over a bar magnet and sprinkle iron filings on it. Draw, before you do it, what you think the pattern will look like. Keep the drawing. Then draw what you think happens with *two* magnets, north facing north.

Do it

1. **One magnet.** Lay the card over a bar magnet. Sprinkle filings thinly and tap the card gently. Compare with your drawing.

2. **Two magnets, attracting.** North facing south, a few centimetres apart. Sprinkle and tap. The lines run straight from one to the other.

3. **Two magnets, repelling.** North facing north. The lines curve away and a region between them stays almost empty.

4. **Walk a compass.** Put a single compass on the card and move it slowly around the magnet, drawing an arrow at each position. Twenty arrows later you have drawn the field yourself, one point at a time — and that is what a field *is*: a direction and a strength at every point in space.

5. **The electric field.** Float grass seed on a thin layer of oil in the dish. Bring a well-charged comb near the surface. The seeds line up along the electric field. It is fainter and fussier than the magnetic version — which is exactly why magnetism was understood first.

What it means

The problem a field solves	The magnet moves the compass without touching it. Either the force jumps across the gap instantly — which turns out to be false — or the magnet changes the <i>space around it</i> , and the compass responds to the space where it sits. The second is true, and the changed space is the field.
Reading a field map	Direction of the line is the direction of the force on a north pole. Where lines crowd together the field is strong; where they spread out it is weak.
Lines never cross	If they did, a compass at the crossing would have to point two ways at once.
Closed loops	Magnetic field lines have no ends — follow one and it returns through the magnet. Electric field lines do have ends: they start on positive charge and finish on negative.
Sharp points concentrate	Field lines crowd at corners and points. This is why a lightning rod is pointed and why a Tesla coil’s topload is a smooth torus — a smooth shape holds more charge before the air breaks down.

Break it

Distance	Move the compass twice as far away. Does the deflection halve, quarter, or something else? You cannot settle this by eye — notice that, and notice that it will need a measurement.
Shielding	Put a sheet of steel between the magnet and the compass, then aluminium, then cardboard. Only one of them changes anything. Why?
Cut it	Break a cheap magnet in half. Do you get a separate north and a separate south, or two smaller complete magnets? Predict first.

The link to the coil
Every part of the coil is a field doing work. The primary makes a magnetic field; the secondary sits in it and responds to it. The capacitor stores energy in an electric field between two plates. The streamer happens when the electric field near the topload gets strong enough to rip air apart. Nothing in the machine touches anything else — it is all fields.

This lesson is low voltage. Everything here runs from batteries or a small plug-in supply and cannot hurt you. That changes at Lesson 10. Build the habits now — one hand, discharge before touching, check before you power up — while the stakes are nothing, so that they are automatic when the stakes are real.

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